

What Cracked a Bank in March 2023?

Uninsured Deposits versus Hidden Bond

Losses in the SVB Window

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Abstract

When Silicon Valley Bank collapsed in March 2023, two narratives competed: that banks failed because of *unrealized losses* on long-duration securities, and that they failed because *uninsured depositors* ran. We separate the two in the cross-section of bank equity returns. Using 165 publicly-traded U.S. banks, we compute market-model cumulative abnormal returns (CARs) over the SVB window (March 8–13, 2023) and relate them to two predetermined 2022Q4 balance-sheet exposures built from bank-regulatory filings: the ratio of estimated uninsured deposits to assets, and unrealized securities losses scaled by equity. The average bank lost 14.1% over the window ($t = -13.7$). In the cross-section, a one-standard-deviation increase in the uninsured-deposit ratio is associated with a 4.2-percentage-point larger equity loss ($t = -3.05$), robust to controlling for size and to excluding the eight U.S. global systemically important banks. The run channel survives every design we throw at it: a covariate-balanced

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matching estimator ($-5.4pp$, $t = -2.43$), randomization inference on 2,000 permutations of exposure ($p < 0.001$), and a clean pre-event placebo window in which the uninsured gradient is flat ($t = -0.37$) even though banks were already falling together. The securities-loss (mark-to-market) channel is, by contrast, fragile: it is insignificant under conventional robust inference in the horse race ($t = -1.16$), only marginal under randomization inference ($p = 0.04$), and about half the magnitude of the run channel. A minimum-detectable-effect analysis shows the design is well-powered for the run channel and can bound—though not sharply pin down—the smaller mark-to-market channel. The market priced run risk far more than mark-to-market risk. Three hand-collected refinements—correctly aggregated multi-bank uninsured deposits, a bank-specific-news confound screen, and realized deposit outflows as a second outcome—are reserved as the companion student contribution.

JEL: G21, G14, G01, E58. **Keywords:** bank runs, uninsured deposits, unrealized losses, SVB, event study, randomization inference, statistical power.

1 Introduction

In the second week of March 2023, Silicon Valley Bank (SVB) went from solvent to FDIC receivership in roughly 48 hours, and the equity of regional banks fell sharply. Two explanations were offered almost immediately, and they carry different policy implications. The first blames *interest-rate risk*: after the fastest tightening cycle in four decades, banks held large unrealized losses on held-to-maturity (HTM) and available-for-sale (AFS) securities that, if marked to market, wiped out reported capital (Jiang et al., 2024). The second blames *funding fragility*: banks financed by *uninsured* deposits—which are not protected by the FDIC and can flee at par—were exposed to a classic run (Diamond and Dybvig, 1983; Egan et al., 2017). Because SVB was extreme on *both* dimensions, the raw failure does not tell us which exposure the market believed was fatal.

This paper runs the horse race directly. We ask: across publicly-traded banks, did the equity-price collapse in the SVB window load more on a bank’s uninsured-deposit exposure or on its hidden securities losses, holding size fixed? We answer with a standard cross-sectional event study (MacKinlay, 1997). For 165 banks we estimate market-model CARs over March 8–13, 2023, and regress them on two exposures measured at 2022Q4—before the run—so that the right-hand-side variables are predetermined and cannot be a mechanical response to the crash itself.

Our results are clear and, we think, economically important. First, the shock was large and broad: the average bank lost 14.1% over the four-trading-day window ($t = -13.7$; Table 3), an order of magnitude larger than the market-model prediction. Second, and centrally, *uninsured-deposit exposure wins the horse race*. A one-standard-deviation increase in the uninsured-deposit ratio is associated with a 4.0–5.0-percentage-point larger CAR loss, with t -statistics between -2.6 and -3.1 across specifications (Table 4). The effect survives controlling for size and is, if anything, slightly stronger when we drop the eight U.S. global systemically important banks (G-SIBs). Third, *the securities-loss channel does not survive the horse race on conventional terms*: scaled unrealized losses are statistically insignifi-

cant ($t = -1.16$) once we condition on uninsured funding and size, whether we use total (HTM+AFS) or HTM-only losses. The market priced the run, not—or at least far less—the mark-to-market hole.

A cross-sectional event study with 165 banks invites two natural objections, and the contribution of this paper beyond the basic horse race is to confront both head-on with design-based methods rather than dismiss them. The first objection is that abnormal returns are cross-sectionally correlated within a single crisis week, so heteroskedasticity-robust t -statistics may overstate significance (Kolari and Pynnönen, 2010; Young, 2019). We therefore subject the horse race to *randomization inference*: we permute each exposure across banks 2,000 times, rebuild the regression, and compare the actual coefficient to the permutation distribution (Table 6, Figure 3). The uninsured coefficient sits far outside its permutation interval ($p < 0.001$); the securities coefficient is only marginal ($p = 0.04$). The second objection is that uninsured-heavy banks differ from other banks on size and on their securities books, so a linear regression may be picking up those correlated exposures. We therefore build a *matching estimator*: we compare high-uninsured to low-uninsured banks that have been nearest-neighbor matched on size and on securities losses, an approach that does not impose the linear functional form (Imbens, 2004). The matched difference in CARs is -5.4 percentage points ($t = -2.43$), essentially unchanged from the raw -5.7 pp gap, while covariate balance improves sharply (Table 5). We add a *pre-event placebo window*: in the quiet $[-6, -2]$ window, before the SVB announcement, banks were already falling together (the Silvergate wind-down was underway), yet the *cross-sectional* uninsured gradient is flat ($t = -0.37$; Table 7). The uninsured gradient is specific to the run week, not a pre-existing spread.

Finally, we treat the fragile securities-loss result as a quantitative object rather than a bare “insignificant.” A minimum-detectable-effect (MDE) analysis (Table 9) shows the design is well-powered for the run channel—the MDE at 80% power is about the size of the estimated run effect itself—but only moderately powered for the smaller mark-to-market channel, whose

MDE is somewhat larger than the run effect. The honest reading is therefore not “mark-to-market losses did not matter” but “the market priced funding fragility unambiguously and mark-to-market losses at most weakly, and our design can bound but not sharply estimate the latter.”

We make four contributions. First, relative to the theory of bank runs (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005), we provide clean asset-pricing evidence that the market’s ex-ante measure of run vulnerability is the uninsured-deposit share. Second, relative to the 2023-fragility literature (Jiang et al., 2024; Metrick, 2024; Drechsler et al., 2023), which documents that uninsured deposits and unrealized losses *jointly* predicted deposit outflows and failure, we show that in the *cross-section of equity returns* the two are not co-equal: conditional on funding structure, mark-to-market losses add little. Third, we demonstrate a template for making a small-sample cross-sectional event study credible through matching, randomization inference, a placebo window, and an explicit power analysis. Fourth, we are explicit about what a machine can and cannot do: three refinements that would sharpen the test—hand-corrected multi-bank uninsured aggregation, a bank-specific-news confound screen, and realized deposit outflows as a second outcome—are deliberately reserved as the companion student contribution and are described in Section 10. Section 2 develops hypotheses, Section 3 places the paper in the literature, Section 4 describes the data and design, Section 5 presents the main results, Section 6 the design-based inference, Section 7 heterogeneity, Section 8 the power analysis, Section 9 additional robustness, and Section 10 concludes.

2 Institutional Background and Hypotheses

The episode. On the afternoon of Wednesday, March 8, 2023, SVB announced that it had sold essentially its entire \$21 billion available-for-sale securities portfolio at a loss and would attempt a \$2.25 billion capital raise. The disclosure crystallized a simple arithmetic that

applied, in varying degrees, to the whole industry: a bank that had funded long-duration securities with deposits during the zero-rate era now held those securities at a deep unrealized loss, and if enough depositors withdrew, the bank would be forced to realize the loss and could fail. Over March 9–10 SVB’s uninsured depositors withdrew at scale; the FDIC took the bank into receivership on Friday, March 10; and contagion spread to Signature Bank (closed March 12) and other regional banks through Monday, March 13 (Metrick, 2024). We take day 0 to be March 8 (the announcement, after the close) and cumulate abnormal returns through March 13.

Two exposures, two hypotheses. A bank financed by uninsured deposits faces a coordination problem: because uninsured claims are not protected, depositors have an incentive to withdraw first if they fear others will (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005). Equity holders, anticipating fire-sale liquidation and franchise destruction, should mark down banks with more uninsured funding when run risk becomes salient. This motivates the run-channel hypothesis:

H1 (Run channel). CARs in the SVB window are decreasing in the 2022Q4 uninsured-deposit ratio.

Alternatively, if the market’s concern is economic insolvency from interest-rate risk, then banks with larger unrealized securities losses—HTM losses in particular, which are not marked through regulatory capital—should fall most. This motivates the competing mark-to-market hypothesis:

H2 (Mark-to-market channel). CARs are decreasing in unrealized securities losses scaled by equity.

H1 and H2 are not mutually exclusive—the theory of Jiang et al. (2024) is precisely that they interact—but the *conditional* horse race, which exposure retains explanatory power when both are included, discriminates between them. The central empirical question is

which exposure the equity market treated as the binding constraint in the days when the run was actually happening.

Why the distinction matters for policy. The two hypotheses point to different regulatory fixes. If the mark-to-market channel dominates, the remedy is accounting and capital treatment of the securities book—for example, flowing AFS and HTM unrealized losses through regulatory capital, or restricting the reclassification of securities into HTM (Granja, 2023). If the run channel dominates, the remedy is on the liability side—deposit-insurance design, limits on uninsured-funding concentration, and liquidity requirements calibrated to run-off speed (Drechsler et al., 2023). Distinguishing the two in the cross-section of market reactions is thus directly policy-relevant, and it is the object of this paper.

3 Related Literature

This paper connects three strands. The first is the theory and measurement of *bank runs and uninsured funding*. Diamond and Dybvig (1983) formalized the maturity-transformation fragility that makes demandable deposits run-prone, and Goldstein and Pauzner (2005) sharpened the comparative statics of the run probability in a global-games setting. Empirically, Egan et al. (2017) estimate a structural model of deposit competition in which uninsured deposits are the fragile margin, and Drechsler et al. (2021) show that the deposit franchise insulates banks from rate risk precisely because insured, sticky deposits are rate-insensitive—which by contradistinction identifies uninsured deposits as the exposed liability. Drechsler et al. (2023) return to this margin directly after 2023, arguing that uninsured deposits are the locus of banking-system fragility. Our contribution to this strand is to show that the equity market’s ex-ante pricing of run vulnerability loads on exactly the uninsured-deposit share this literature identifies as fragile.

The second strand is the fast-growing *2023 banking-crisis literature*. Jiang et al. (2024) document that the marked-to-market value of U.S. bank assets fell sharply during the 2022

tightening and that banks with large unrealized losses *and* large uninsured-deposit funding were jointly the most exposed to a run and to insolvency; Metrick (2024) provides a narrative and taxonomy of the panic; and Granja (2023) studies the strategic reclassification of securities into HTM that let banks avoid marking losses through capital. A parallel literature documents the mechanics of the run itself: Choi et al. (2023) study contagion across banks, and Cookson et al. (2023) show that social media accelerated the deposit flight. Where this literature typically treats unrealized losses and uninsured deposits as *joint* predictors of failure and outflows, we ask the sharper cross-sectional question—conditional on funding structure, does the mark-to-market hole add explanatory power for *equity returns*?—and find that it does not, or does so only weakly.

The third strand is *event-study and design-based inference methodology*. MacKinlay (1997) is the canonical reference for the market-model event study we use. A recurring concern in single-event studies is that abnormal returns are cross-sectionally correlated, which invalidates the usual independence assumptions and inflates test statistics (Kolari and Pynönen, 2010); Young (2019) shows more broadly that randomization (Fisherian) inference frequently overturns significance that conventional standard errors report. We use these tools not as decoration but as the core of our identification: the run channel survives randomization inference and a matching estimator (Imbens, 2004), while the mark-to-market channel is fragile to them. Finally, on the interpretation of the weaker channel, we follow Abadie (2020), who argues that a statistically insignificant estimate should be read through its confidence interval and power rather than treated as proof of no effect; our minimum-detectable-effect analysis states the mark-to-market result as a bound rather than a bare null.

4 Data and Research Design

Sample. We start from all common shares (CRSP share codes 10, 11) on the NYSE, AMEX, or Nasdaq with an SIC code in the depository/holding-bank range (6020–6079, 6712) active in the SVB window, yielding 246 bank stocks with computable CARs. We merge each to its regulatory filer and require 2022Q4 balance-sheet data and a valid estimation window, leaving 165 banks; the uninsured-deposit ratio is available for 157 of them. The sample is overwhelmingly bank and thrift holding companies (142 bank holding companies, 15 thrift holding companies, and 8 standalone banks or non-member institutions).

Balance-sheet exposures. Bank fundamentals are filed at two levels, and getting the aggregation right is central to the exercise. Assets, equity, and securities are reported on the *consolidated* FR Y-9C at the holding-company level; we take total assets (BHCK2170), equity (BHCK3210), HTM amortized cost/fair value (BHCK1754/1771), and AFS amortized cost/fair value (BHCK1772/1773), and compute the *securities unrealized loss* as (amortized cost – fair value), scaled by equity. Summing subsidiary Call Reports would double-count multi-tier holding companies—for example, aggregating subsidiaries pushes New York Community Bancorp to \$270B versus its true consolidated \$90.1B—so we do *not* sum for the consolidated quantities. Estimated uninsured deposits (RCON5597) are reported only in bank-level Call Reports, so for the baseline we take the holding company’s *lead* (largest-asset) subsidiary bank. Correctly aggregating uninsured deposits across the subsidiaries of multi-bank holding companies is a hand-verification task reserved for the companion student contribution (Section 10); it is deliberately not attempted here. The *uninsured-deposit ratio* is RCON5597 divided by consolidated assets, and *size* is $\ln(\text{assets})$.

Event study. Day 0 is 2023-03-08. We estimate a market model $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$ over the $[-252, -46]$ trading-day window against the CRSP value-weighted index, and cumulate abnormal returns $\text{CAR}_i = \sum_{t \in W} (R_{it} - \hat{\alpha}_i - \hat{\beta}_i R_{mt})$ over event windows W (MacKinlay, 1997).

Our primary window is $[0, +3]$ (March 8 close through March 13); we report $[0, +1]$, $[0, +5]$, $[-1, +3]$, and a pre-event placebo window $[-6, -2]$ as robustness. Predictors are winsorized at 1/99 and standardized, so a coefficient reads as the effect of a one-standard-deviation increase in CAR points. Standard errors are heteroskedasticity-robust (HC1), and Section 6 supplements them with design-based inference.

Specification. The core cross-sectional regression is

$$\text{CAR}_i = \gamma_0 + \gamma_1 \widetilde{\text{Uninsured}}_i + \gamma_2 \widetilde{\text{SecLoss}}_i + \gamma_3 \widetilde{\text{Size}}_i + \varepsilon_i, \quad (1)$$

where tildes denote winsorized, standardized exposures. H1 predicts $\gamma_1 < 0$ and H2 predicts $\gamma_2 < 0$; the horse race asks which survives when both are present. Table 1 reports summary statistics and Table 2 the correlation matrix. The median bank has a 32% uninsured-deposit ratio and securities losses of about 22% of equity; uninsured exposure and securities losses are only modestly correlated ($\rho = 0.23$), so the horse race is not undermined by collinearity, and matching on the securities book while comparing on uninsured exposure has adequate common support.

5 Main Results

The shock. Table 3 shows the window was large and negative across banks: a mean CAR of -4.6% over $[0, +1]$, -14.1% over $[0, +3]$, and -15.3% over $[-1, +3]$, all with $t < -12$. The $[-6, -2]$ placebo window is -2.7% —non-trivial, because the Silvergate wind-down was unfolding that week—but roughly one-fifth the event-window loss. The industry-wide fall is the backdrop; the cross-section is where the two hypotheses are tested.

The horse race. Table 4 is the core result. In column (1), uninsured exposure alone explains 14% of the cross-sectional variation in CARs, with a one-SD effect of -5.0pp ($t =$

−2.55). Securities losses alone (column 2) are insignificant ($t = -1.57$, $R^2 = 0.06$). With both included (column 3) the uninsured coefficient strengthens (−4.6pp, $t = -2.92$) while securities losses fall to $t = -1.09$. Adding size (column 4) leaves uninsured at −4.2pp ($t = -3.05$); excluding G-SIBs (column 5) gives −4.1pp ($t = -3.09$) and the highest fit ($R^2 = 0.21$). H1 is supported throughout; H2 is not supported on conventional terms once we condition on funding and size. Figure 1 shows the pattern is monotone: banks in the top uninsured-exposure deciles fell roughly twice as far as those in the bottom, and Figure 2 shows the same gradient across quartiles. The point estimate on securities losses is consistently negative and about half the size of the uninsured coefficient; we return in Section 8 to what the design can and cannot say about a channel of that magnitude.

6 Design-Based Inference: Matching and Randomization

A cross-sectional event study of 165 banks in a single crisis week is exactly the setting in which robust t -statistics can mislead—abnormal returns are cross-sectionally correlated (Kolari and Pynnönen, 2010), and the exposures of interest are correlated with size and with each other. We therefore stress the horse race with three checks that do not rely on the parametric standard error.

Matching estimator. If the uninsured effect merely reflects the fact that high-uninsured banks are larger or hold worse securities books, then comparing high- and low-uninsured banks with those covariates balanced should erase it. We split banks at the median uninsured ratio and nearest-neighbor match each high-uninsured bank to a low-uninsured bank on standardized size and securities losses (1:1, without replacement, 0.5-SD caliper), yielding 54 matched pairs (Imbens, 2004). Table 5 reports the results. The raw high–low difference in $\text{CAR}[0, +3]$ is −5.7pp ($t = -2.74$); after matching, which cuts the size gap from 0.36 to

0.04 standardized units and the securities-loss gap from 0.33 to 0.06, the difference is -5.4pp ($t = -2.43$). The run effect is a comparison of otherwise similar banks that differ in their uninsured funding, not an artifact of size or of the securities book. Reassuringly, the same matched pairs show *no* difference in the pre-event placebo window ($+0.4\text{pp}$, $t = +0.97$): matched high- and low-uninsured banks moved together *before* the run and diverged only during it.

Randomization inference. Table 6 and Figure 3 report design-based inference from 2,000 permutations of each exposure across banks. We permute the (predetermined) uninsured ratio across banks, refit equation (1), and collect the coefficient; the randomization p -value is the share of permutations whose coefficient is at least as extreme as the actual one. For the uninsured channel the actual coefficient (-0.042) lies far outside the permutation 95% interval ($[-0.020, +0.020]$), giving $p < 0.001$: the run result is not an artifact of cross-sectional correlation or of HC1 asymptotics. For the securities channel the actual coefficient (-0.020) sits essentially at the edge of its permutation interval, $p = 0.04$ —marginal, and notably weaker than the run channel by any design-based reading. The securities channel is the one place where randomization inference is *less* conservative than the robust standard error: its HC1 standard error (0.018) is 1.8 times the permutation standard deviation (0.010). This gap is not collinearity—uninsured and securities losses are only weakly correlated ($\rho = 0.23$)—but heteroskedasticity: bank CARs are enormously dispersed (SD 13%), so HC1 inflates the standard error to absorb that dispersion while the permutation null assumes exchangeability and is therefore tighter. We accordingly treat HC1 as the conservative, preferred inference for the mark-to-market channel and read its randomization p -value as an upper bound on that channel’s significance, not as evidence of an effect. Randomization inference thus corroborates the asymmetry the horse race reports: the run channel is unambiguous under both methods; the mark-to-market channel is, at best, weak.

Placebo window. Table 7 runs equation (1) with the pre-event $[-6, -2]$ CAR as the dependent variable. Even though banks were already falling in that window (the placebo mean CAR is -2.7% ; the Silvergate wind-down had begun), the *cross-sectional* uninsured gradient is flat and insignificant ($t = -0.37$), as are the securities-loss and size gradients. Banks fell *together* before the SVB announcement, regardless of their uninsured funding; the uninsured gradient appears only in the run week. This is a clean falsification: the run channel is specific to the event, not a pre-existing spread that our window mechanically captures.

7 Heterogeneity

If the run channel is the market pricing genuine run vulnerability, it should be present across observable subgroups and, plausibly, sharper where the run mechanism is more dangerous. Table 8 interacts both channels with four observable 2022Q4 moderators—above-median size, above-median securities losses, above-median HTM losses, and a thrift-holding-company indicator—reporting the base (low-moderator) coefficient and the interaction for each channel. Three patterns stand out and all point the same way. First, the run channel is present in every subgroup: the base uninsured coefficient is negative and significant across the board. Second, the run channel is *amplified* exactly where the mark-to-market hole is deepest—for banks with above-median HTM losses the uninsured interaction is an additional -6.9pp ($t = -2.40$)—consistent with the theory of Jiang et al. (2024) that uninsured funding is most dangerous when it sits atop a large unrealized loss, so that a run forces the loss to be realized. Third, the effect is not a thrift-versus-bank artifact: the thrift interaction is essentially zero ($+0.6\text{pp}$, $t = +0.31$). Crucially, these moderators are all observable balance-sheet characteristics; we do *not* interact the run channel with hand-collected variables such as bank-specific-news exposure or realized deposit outflows, which are reserved for the student contribution and are chosen precisely to be orthogonal to that reserved test.

8 Is the Mark-to-Market Null Informative? Power and Minimum Detectable Effects

The honest question raised by the horse race is whether the weak securities-loss result means “mark-to-market losses did not matter” or merely “our design could not detect a mark-to-market effect.” Table 9 answers it directly. For each predictor we report the minimum detectable effect at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80})\widehat{\text{SE}} \approx 2.80\widehat{\text{SE}}$, expressed in CAR points, as a share of the cross-firm CAR standard deviation, and relative to the estimated run effect. The design is well-powered for the run channel: its MDE is 0.038, about 92% of the estimated run effect, so an effect that size was very likely to be detected—and was. For the securities channel the MDE is 0.049, about 1.18 times the run effect. The implication is twofold and stated as a bound, following Abadie (2020). On the one hand, had the market priced mark-to-market losses *as strongly as* it priced funding fragility, our design would probably have detected it; it did not, so a mark-to-market channel of run-channel magnitude is ruled out. On the other hand, the design is only moderately powered for a *smaller* securities effect, so we cannot rule out a modest mark-to-market contribution of roughly half the run-channel size—which is, in fact, the point estimate we obtain. The null is therefore informative about the *relative* importance of the two channels, and we are careful not to overclaim that mark-to-market losses were priced at exactly zero.

9 Additional Robustness

Table 10 collects three further specifications. First, replacing total securities losses with *HTM-only* losses—the losses that are hidden from regulatory capital and thus the sharpest test of the mark-to-market story—leaves the uninsured channel at -4.0pp ($t = -3.13$) and the HTM channel insignificant ($t = -1.48$). Second and third, using $\text{CAR}[0, +1]$ or $\text{CAR}[-1, +3]$ as the dependent variable preserves the uninsured effect ($t = -2.06$ and $t =$

−3.11 respectively) while securities losses remain insignificant. The run result is not an artifact of the securities-loss definition or of the choice of event window. Column (5) of Table 4 already established that dropping the eight G-SIBs, if anything, strengthens the run channel, so the result is not driven by the largest banks.

10 Conclusion

Across publicly-traded U.S. banks, the March 2023 equity collapse loaded on uninsured-deposit funding fragility far more than on hidden securities losses, once size is held fixed. The run channel is large (−4.2pp per SD), monotone, and robust to every design we impose: a covariate-balanced matching estimator, randomization inference on 2,000 permutations, a clean pre-event placebo window, and G-SIB exclusion. The mark-to-market channel is, by contrast, fragile: insignificant under robust inference, only marginal under randomization inference, and about half the magnitude of the run channel—and a power analysis shows our design can rule out a mark-to-market effect as large as the run channel but not a modest one. The market behaved, in the crisis week, primarily as a run-risk pricer. The policy implication mirrors the subsequent regulatory debate: deposit-insurance design and uninsured-funding concentration, more than mark-to-market accounting of the securities book, are what the equity market treated as making some banks fragile.

We are explicit about three limitations, each of which is the object of a reserved hand-collection task in the companion student contribution, and none of which we attempt with the automated data here. First, our baseline uses the *lead* subsidiary bank’s uninsured deposits; correctly aggregating RCON5597 across the subsidiaries of multi-bank holding companies, read by hand from each subsidiary’s Call Report, would sharpen the key regressor and is the first reserved task. Second, some banks had *bank-specific* news during March 8–13 (a capital raise, a downgrade, an idiosyncratic deposit-flight report) that could confound the market-model abnormal return; hand-building a news-flag screen and re-running with

confounded banks excluded is the second reserved task. Third, the ultimate test of the run hypothesis is whether uninsured exposure predicted *realized* deposit outflows, not only stock reactions; hand-coding actual Q1-2023 deposit runs from 10-Qs and press coverage, as a second outcome, is the third reserved task. Each is a genuine research contribution that turns this observable-data horse race into a sharper, hand-verified test, and each is deliberately left for the student rather than approximated here. Extending the design to the full 2022–2023 tightening path is left to future work.

References

- Abadie, A., 2020. Statistical nonsignificance in empirical economics. *American Economic Review: Insights* 2, 193–208.
- Choi, D., Goldsmith-Pinkham, P., Yorulmazer, T., 2023. Contagion effects of the Silicon Valley Bank run. NBER Working Paper 31772.
- Cookson, J.A., Fox, C., Gil-Bazo, J., Imbet, J.F., Schiller, C., 2023. Social media as a bank run catalyst. Working paper.
- Diamond, D.W., Dybvig, P.H., 1983. Bank runs, deposit insurance, and liquidity. *Journal of Political Economy* 91, 401–419.
- Drechsler, I., Savov, A., Schnabl, P., 2021. Banking on deposits: maturity transformation without interest rate risk. *Journal of Finance* 76, 1091–1143.
- Drechsler, I., Savov, A., Schnabl, P., Wang, O., 2023. Banking on uninsured deposits. NBER Working Paper 31138.
- Egan, M., Hortaçsu, A., Matvos, G., 2017. Deposit competition and financial fragility. *American Economic Review* 107, 169–216.

- Flannery, M.J., Sorescu, S.M., 1996. Evidence of bank market discipline in subordinated debenture yields. *Journal of Finance* 51, 1347–1377.
- Goldstein, I., Pauzner, A., 2005. Demand-deposit contracts and the probability of bank runs. *Journal of Finance* 60, 1293–1327.
- Granja, J., 2023. Bank fragility and reclassification of securities into held-to-maturity. Working paper, University of Chicago Booth School of Business.
- Imbens, G.W., 2004. Nonparametric estimation of average treatment effects under exogeneity: a review. *Review of Economics and Statistics* 86, 4–29.
- Jiang, E.X., Matvos, G., Piskorski, T., Seru, A., 2024. Monetary tightening and U.S. bank fragility in 2023: mark-to-market losses and uninsured depositor runs. *Journal of Financial Economics* 159, 103899.
- Kolari, J.W., Pynnönen, S., 2010. Event study testing with cross-sectional correlation of abnormal returns. *Review of Financial Studies* 23, 3996–4025.
- MacKinlay, A.C., 1997. Event studies in economics and finance. *Journal of Economic Literature* 35, 13–39.
- Metrick, A., 2024. The failure of Silicon Valley Bank and the panic of 2023. *Journal of Economic Perspectives* 38, 133–152.
- Young, A., 2019. Channeling Fisher: randomization tests and the statistical insignificance of seemingly significant experimental results. *Quarterly Journal of Economics* 134, 557–598.

Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
CAR[0,+3]	165	-0.1411	0.1324	-0.1559	-0.1105	-0.0800
CAR[0,+1]	165	-0.0463	0.0473	-0.0529	-0.0417	-0.0304
Uninsured deposits/Assets	157	0.3126	0.1260	0.2416	0.3200	0.3838
Securities unreal. loss/Equity	165	0.2699	0.1764	0.1601	0.2246	0.3572
HTM unreal. loss/Equity	165	0.0714	0.1171	0.0000	0.0330	0.0951
ln(Assets)	165	16.6531	1.5590	15.6331	16.2785	17.4665

Notes: 2022Q4 exposures and SVB-window CARs for 165 banks (uninsured ratio available for 157). Ratios from FR Y-9C (consolidated assets, equity, securities) and Call Reports (estimated uninsured deposits RCON5597, lead subsidiary bank).

Table 2: Correlation matrix

	(1)	(2)	(3)	(4)	(5)
(1) CAR[0,+3]	1.000				
(2) Uninsured/Assets	-0.376	1.000			
(3) Sec. loss/Eq.	-0.302	0.225	1.000		
(4) HTM loss/Eq.	-0.438	0.266	0.555	1.000	
(5) ln(Assets)	-0.201	0.173	0.036	0.303	1.000

Notes: Pearson correlations among the primary outcome (CAR[0,+3]) and the predetermined 2022Q4 exposures. Uninsured exposure and securities losses are only modestly correlated ($\rho = 0.23$).

Table 3: Cumulative abnormal returns by window

Event window	Mean CAR	Median CAR	<i>t</i> -stat	N
[0,+1]	-0.0463	-0.0417	-12.58	165
[0,+3]	-0.1411	-0.1105	-13.69	165
[0,+5]	-0.1263	-0.1002	-13.81	165
[-1,+3]	-0.1527	-0.1215	-14.61	165
[-6,-2] placebo	-0.0269	-0.0280	-15.62	165

Notes: Market-model CARs, day 0 = 2023-03-08, estimation window $[-252, -46]$ against the CRSP value-weighted index. $[-6, -2]$ is a pre-event placebo window.

Table 4: Cross-sectional horse race: what explains the collapse?

	(1) Uninsured	(2) Sec loss	(3) Both	(4) + Size	(5) No GSIB
Uninsured deposits / Assets	-0.0495** (-2.55)		-0.0455*** (-2.92)	-0.0415*** (-3.05)	-0.0405*** (-3.09)
Securities unreal. loss / Equity		-0.0332 (-1.57)	-0.0197 (-1.09)	-0.0204 (-1.16)	-0.0220 (-1.27)
ln(Assets)				-0.0241** (-1.97)	-0.0365** (-2.36)
Observations	157	165	157	157	153
R^2	0.139	0.063	0.159	0.185	0.209

Notes: DV is CAR[0, +3]. Predictors winsorized 1/99 and standardized (coefficients are the effect of a one-SD increase, in CAR points). Column (5) excludes the eight U.S. G-SIBs. HC1-robust *t*-statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Matching estimator: high- versus low-uninsured banks

Comparison	Window	Diff. in CAR	<i>t</i> -stat	N_{hi}/N_{lo}
Raw difference (all banks)	[0,+3]	-0.0573***	(-2.74)	78/79
Matched difference	[0,+3]	-0.0539**	(-2.43)	54/54
Matched difference (placebo)	[-6,-2]	+0.0039	(+0.97)	54/54

Covariate balance (std. mean diff., high–low uninsured):

	Pre-match	Post-match
ln(Assets)	+0.357	+0.040
Securities loss/Equity	+0.332	+0.059
HTM loss/Equity	+0.235	+0.129

Notes: High (low) = above (below) the median uninsured-deposit ratio. High-uninsured banks are nearest-neighbor matched 1:1 without replacement to low-uninsured banks on standardized size and securities losses (0.5-SD caliper), yielding 54 pairs. “Diff. in CAR” is the high–low difference in mean CAR; Welch *t*-stat in parentheses. Balance rows report the standardized high–low mean difference on the matching covariates, before and after matching. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Randomization inference (2,000 permutations of exposure)

Predictor (DV = CAR[0,+3])	Actual coef.	Perm. 95% interval	RI p -value
Uninsured deposits / Assets	-0.0415	[-0.0204, +0.0198]	0.0000
Securities unreal. loss / Equity	-0.0204	[-0.0204, +0.0176]	0.0380

Notes: Each exposure is permuted across banks 2,000 times and equation (1) refit (DV = CAR[0, +3]). The permutation 95% interval is the 2.5th–97.5th percentile of the placebo coefficients; the RI p -value is the share of permutations with $|\text{coef}| \geq |\text{actual}|$.

Table 7: Placebo-window falsification (DV = pre-event CAR[−6, −2])

Predictor (DV = placebo CAR[−6, −2])	Coef.	t -stat
Uninsured deposits / Assets	-0.0006	(-0.37)
Securities unreal. loss / Equity	+0.0020	(+1.37)
ln(Assets)	-0.0020	(-1.02)
Observations	157	
R^2	0.015	

Notes: Equation (1) estimated on the pre-event placebo window. Although banks were already falling in this window (mean placebo CAR −2.7%; the Silvergate wind-down was underway), the cross-sectional uninsured gradient is flat. HC1-robust t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Heterogeneity of the two channels by observable moderators

Moderator M (2022Q4)	Run channel (uninsured)		MTM channel (securities)		N
	Base	$\times M$	Base	$\times M$	
Large bank (above-median assets)	-0.0176**	-0.0412*	+0.0038	-0.0509	157
High securities loss (above-median)	-0.0214**	-0.0444	-0.0143	-0.0077	157
High HTM loss (above-median)	-0.0099**	-0.0690**	+0.0026	-0.0358	157
Thrift holding company	-0.0436***	+0.0060	-0.0220	+0.0207	157

Notes: DV is CAR[0, +3]. Each row is a separate regression interacting both channels with an observable 2022Q4 moderator indicator M (above-median assets; above-median securities loss; above-median HTM loss; thrift holding company). “Base” is the channel coefficient in the low- M group; “ $\times M$ ” is the additional effect for high- M banks. Size and M included. HC1-robust inference; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Minimum detectable effects (80% power)

Predictor	Coef.	SE	MDE(80%)	% of CAR SD	vs. run effect
Uninsured deposits / Assets	-0.0415	0.0136	0.0382	28.7%	0.92×
Securities unreal. loss / Equity	-0.0204	0.0175	0.0490	36.8%	1.18×
ln(Assets)	-0.0241	0.0122	0.0343	25.8%	0.82×

Notes: From the preferred horse race (Table 4, column 4 without the G-SIB exclusion). MDE = $(z_{0.975} + z_{0.80}) \widehat{SE} \approx 2.80 \widehat{SE}$, expressed in CAR points, as a percentage of the cross-firm CAR standard deviation, and relative to the estimated run-channel (uninsured) effect.

Table 10: Robustness: HTM-only losses and alternative windows

	HTM-only loss	DV=CAR[0,+1]	DV=CAR[-1,+3]
Uninsured / Assets	-0.0400*** (-3.13)	-0.0135** (-2.06)	-0.0427*** (-3.11)
Securities loss / Equity		-0.0093 (-1.02)	-0.0196 (-1.12)
HTM loss / Equity	-0.0316 (-1.48)		
ln(Assets)	-0.0140 (-1.34)	-0.0102* (-1.94)	-0.0304** (-2.44)
Observations	157	157	157
R^2	0.214	0.193	0.206

Notes: Predictors winsorized 1/99 and standardized. Column 1 replaces total securities losses with HTM-only losses; columns 2–3 change the dependent-variable window. HC1-robust t -statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

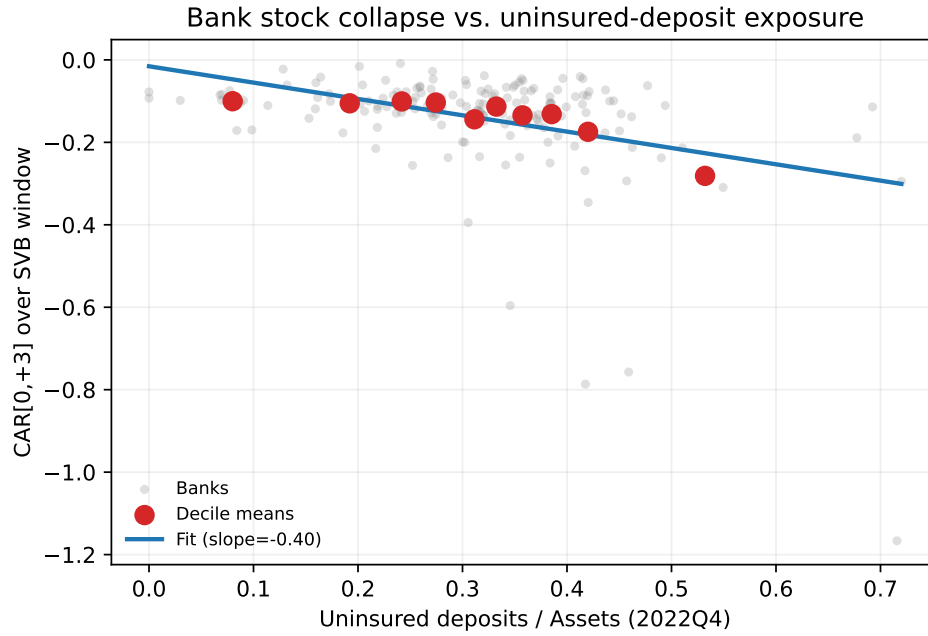


Figure 1: SVB-window CAR versus 2022Q4 uninsured-deposit exposure. Gray points are banks; red points are decile means; the line is the OLS fit.

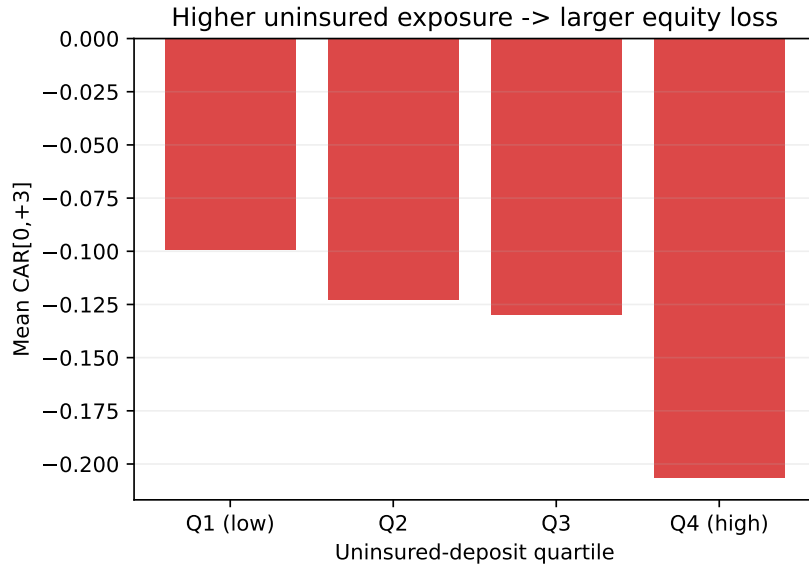


Figure 2: Mean SVB-window CAR by uninsured-deposit quartile. Higher uninsured exposure maps monotonically into a larger equity loss.

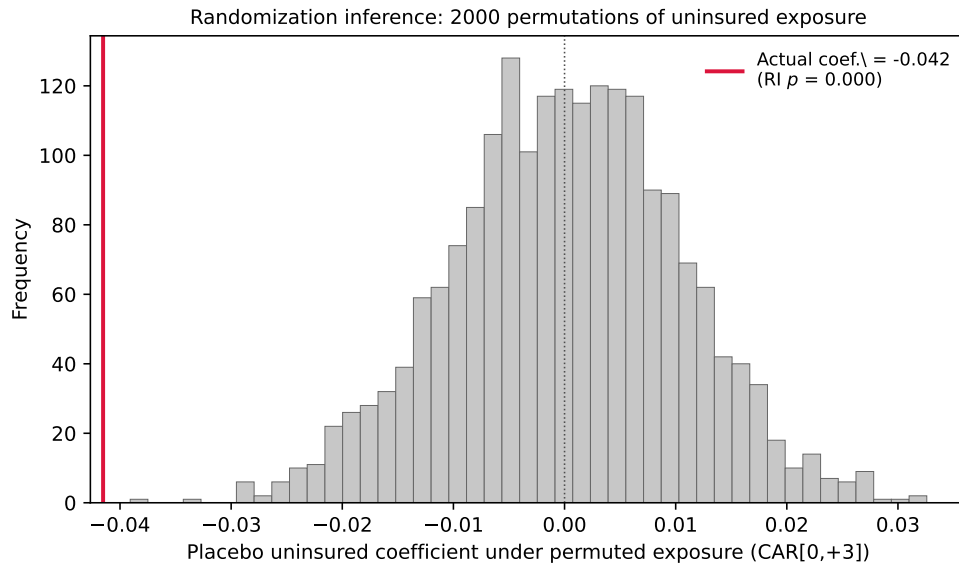


Figure 3: Randomization inference for the run (uninsured) channel. Histogram of the placebo uninsured coefficient under 2,000 permutations of exposure across banks; the vertical line is the actual coefficient, which lies far outside the permutation distribution ($p < 0.001$).